

An Introduction to the Specified Pool Sector

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- ▶ Although a significant portion of trading in agency MBS pass-throughs occurs in the to-be-announced (TBA) market, the importance of the specified pool market has grown rapidly since the beginning of 2003. In the specified pool market, investors pay a premium over TBAs either to take delivery of desirable pools or to simply reduce the risk of receiving adverse collateral in the TBA market. In this primer, we discuss some important aspects of the specified pool market.
- ▶ **Seasoning:** Prepay speeds are slow immediately after the mortgage is financed and gradually increase over a period of time that depends on several economic factors and borrower characteristics. Typically, seasoned pools of discount coupon and new origination pools of premium coupon pass-throughs command a pay-up over their TBA counterparts. Seasoned pools trade actively in the MBS market because their pay-ups are included in different MBS benchmark indices.
- ▶ **Loan Size:** Loan size has a substantial impact on the prepayment characteristics of a mortgage. The loan size-based segment of the specified pool market may be subdivided into four groups: low loan balance (LLB), moderate loan balance (MLB), high moderate loan balance (HLB) and generic pools.
- ▶ **Geography:** The geographic location in which a mortgage has been originated plays an important role in determining prepay speeds. Usually, slower than average prepayment speeds in some states are due to certain state regulations which increase the transaction costs involved with taking out a mortgage. Mortgage pools with a concentration of loans from “slow” prepaying states command a pay-up over TBAs for premium pass-throughs.
- ▶ **Leveraged Borrowers/Loans:** Low FICO and high LTV pools are supposed to offer both extension and call protection versus generic (TBA deliverable) collateral. At the peak of the Refi wave in 2003, both low FICO and high LTV pools offered significant prepayment protection but this protection withered away as the Refi wave ended. In general, the level of prepayment and extension protection these pools provide depends on several factors. For instance, in the past, high LTV pools have appeared to provide excellent extension protection but it is not clear if the same protection would be available in a strongly negative HPA scenario.
- ▶ Most of our discussion about fair pay-ups for specified pool collateral proceeds in terms of equal OAS pay-ups. The pay-ups depend on the “in-the-moneyness” of the relevant security, the specialness of dollar rolls, demand from the CMO market, the shape of the yield curve and the market’s expectations for future rate movements among other factors. In general, specified pools rarely trade at more than 50% of their equal OAS-implied theoretical value.

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I. Introduction

Although a significant portion of trading in agency MBS pass-throughs occurs in the to-be-announced (TBA) market, the importance of the specified pool market has grown rapidly since the beginning of 2003. The TBA market is a forward delivery market with participants entering into forward contracts to buy or sell MBS on a monthly settlement schedule set by the Bond Market Association. The pools delivered by the seller on the settlement date have to satisfy Good Delivery Guidelines but within these good delivery guidelines, sellers of TBAs have a lot of discretion over what pools they can deliver to the buyer. Thus, the buyer of a pass-through security in the TBA market does not know in advance what pools they are going to receive until the pool notification day, whereas buyers in the specified pool market know exactly what pools they are buying at the time of the trade itself.

Several factors have contributed to the rapid growth of the specified pool market over the past 4-5 years:

1. In 2003, several buyers of premium pass-through securities in the TBA market were delivered mortgage pools that prepaid substantially faster than the average speeds for the corresponding cohort. As investors suffered losses on their premium pass-through holdings because of this adverse delivery feature of TBAs, the specified pool market took off as an alternative to the TBA market.
2. At about the same time that interest rates were at generational lows in 2003, Fannie Mae and Freddie Mac started releasing lot of additional pool-level information on agency pools. The availability of this information allowed investors to analyze the prepayment characteristics of agency pools with respect to several attributes which further contributed to a rise in the number of variables that are actively priced in the specified pool market.
3. The number of CMO deals using “story collateral” (i.e., collateral with desirable prepayment characteristics) has picked up recently which has further contributed to the increased importance of the specified pool market.

In the specified pool market, investors pay a premium over equal coupon TBAs either to take delivery of pools that have some desirable prepayment characteristics or to simply reduce the risk of receiving adverse collateral in the TBA market. The premium or pay-up that investors are willing to pay for a specified pool depends on several factors including the coupon of the underlying security, the call/extension protection offered by the pool, the risks that the market is focused on at that point of time, among other factors. Pool-level characteristics that are very desirable to CMO desks usually command a high premium over TBAs. Finally, the specialness of the dollar roll of the underlying coupon TBA plays a very important role in determining the pay-ups commanded by specified pools. This primer provides a brief introduction to the collateral characteristics that are most actively priced and traded in the agency specified pool market.

II. Actively Traded Pool-level Characteristics in the Specified Pool Sector

Seasoning

Typically, prepayment speeds are relatively slow immediately after a mortgage is originated and gradually increase over a period of time in a manner that depends on several economic variables and borrower characteristics.¹ The **seasoning ramp** for mortgages captures the fact that most households get progressively more “dissatisfied” with their housing as their income trends up and the size of their family increases. In addition, the sunk costs associated with the initial move become less of a factor over time.

Investors in discount pass-throughs prefer to receive principal on their bonds sooner rather than later because these bonds offer a below-market coupon rate. An investor can satisfy this objective by accumulating seasoned bonds since these pools have shorter maturities and faster prepayment speeds compared to newly originated pools. Thus, investors should be willing to pay a premium over TBAs for seasoned discount pools. The opposite occurs with premium pass-throughs as owners of premium coupon pools prefer their principal balances to remain outstanding for a longer period. Consequently, newly originated premium pools trade at a premium versus their TBA counterparts.² In general, the term “seasoned pools” refer to seasoned discount pass-throughs, while newly originated premium pools are referred to as “low WALA” or “new origination pools.” Seasoned pools are very actively traded in the MBS market because pay-ups for seasoning are usually included in different MBS benchmark indices, which is unlike the case with other specified pool characteristics.

The pay-ups for seasoned discount pools depend on the following factors:

- **The “out-of-the-moneyness” of the TBA:** The higher the discount, the larger the pay-up for seasoning. This makes intuitive sense as faster prepayment speeds and shorter maturities are worth more relative to TBAs when the underlying pass-through is at a deeper discount.
- **Shape of the yield curve:** Steeper yield curves usually lead to higher pay-ups for seasoned pools. A steep yield curve offers the added advantage of roll down along the curve to seasoned pools making them worth more relative to TBAs than in a flat yield curve scenario.
- **Dollar rolls:** Pay-ups for seasoning decline when dollar roll of the underlying TBA trades special. For instance, in March 2007, there was a 3.5-4.0 tick pay-up for 2005 origination FNMA 5s but the pay-up for the same pools dropped to 0 ticks in early April 2007 when the April/May dollar roll of FN 5s traded 4 ticks special. This is not necessarily bad for the specified pool position depending upon how it has been hedged. If the pool has been hedged with Treasuries, swaps or agencies, the position does not necessarily lose because the TBA itself is likely to trade rich as the dollar roll gets special. If the pool is hedged with TBAs, the specified pool holder essentially loses the amount the pay-up drops by as the dollar roll starts to trade special (on a mark-to-market basis).

¹ See our primer titled *Residential Mortgages: Prepayments and Prepayment Modeling* for more details.

² It is worth pointing out here that the seasoning ramp most relevant for premium passthroughs is the ramp for refinancings, while the one relevant for discounts is the ramp for housing turnover.

- **Demand from the Agency CMO market:** The CMO sector allows investors to express a leveraged view on prepayments. Investors who like the collateral characteristics of a specific vintage may drive up pay-ups. However, there are also several instances when CMO desks actually drive up dollar rolls of discount coupons with the expectation of receiving more seasoned collateral in the TBA delivery process. In those cases, although pay-ups for seasoning actually decline due to demand for the collateral from the CMO desks, owners of specified pools do not necessarily lose because the TBA itself is likely to trade quite rich thus reflecting the potential for seasoned pools being delivered into the TBA.
- **Vintage Characteristics:** Sometimes differences between collateral characteristics of different vintages (other than age) also impact pay-ups for seasoning. As an example, several investors expect the 2004 cohort to show faster turnover speeds than 2006 cohort (adjusted for age) due to the high HPA experienced by the 2004 cohort.

Figure 1 shows theoretical and actual pay-ups for different 30-year seasoned discount pools. It is worth noting that while the pay-ups shown here are quoted at the origination year level, in practice, seasoned pool trading may take place in terms of WALA (the seasoning is expressed in months as opposed to years). The theoretical pay-ups in the figure are calculated by setting the OAS of seasoned pools equal to the OAS of the TBA with the corresponding coupon. Typically, seasoned pools trade at 30%-50% of their equal OAS payups when the dollar roll of the underlying coupon is not special.

Figure 1: Origination Year Pay-ups for Seasoned Collateral

FNCL 5: OAS = -7.8					
Orig. Year	Average Loan Size	Price	Equal OAS Price over TBA (ticks)	Market Pay-up (ticks)	(Market Pay-up)/(Equal OAS Pay-up)
2003	\$161K	97-04	26.0	8.5	33%
2004	\$178K	97-00	22.0	7.0	32%
2005	\$196K	96-25	15.0	5.0	33%
2006	\$217K	96-18+	8.5	1.0	12%
TBA	\$220K	96-10			

FNCL 5.5: OAS = -6.0					
Orig. Year	Average Loan Size	Price	Equal OAS Price over TBA (ticks)	Market Pay-up (ticks)	Market Pay-up/Equal OAS
2003	\$144K	99-09	19.0	8.0	42%
2004	\$154K	99-05+	15.5	6.0	39%
2005	\$166K	99-00+	10.5	4.5	43%
2006	\$212K	98-25	3.0	0.8	25%
TBA	\$230K	98-22			

As of 5/11/2007

Source: Banc of America Securities

Most specified pool investors supplement the OAS-based analysis presented above with some carry based analysis. The following example illustrates the basic idea behind this

carry analysis. For the month of June 2007, our model projects a prepayment speed of 11% CPR for 2004 FN 5s and 2% CPR for TBA 5s (2 WALA collateral). The 9% CPR differential between these prepayment speeds lead to approximately 1.06 ticks per month of higher carry on the 2004 FN 5s at current dollar price levels (as of 05/21/2007 closes). However, it should be noted that the carry advantage of seasoned pools declines when the dollar roll of the underlying TBA is special (i.e., TBAs offer better carry than implied by prepayment speeds on the TBA deliverable collateral).

From the perspective of hedging, there are some differences between duration, curve shape and convexity exposures of seasoned pools and equal coupon TBAs. For instance, seasoned discount pools have shorter durations, higher steepener exposures and better convexities than corresponding TBAs. These differences are frequently ignored by investors but can be very important when interest rates or curve shape change significantly.

Loan Sizes: LLB/MLB/HLB Pools

It is well recognized in the market that loan size has a significant impact on the prepayment characteristics of mortgage pools. During the Refi wave of 2002-2004, an active specified pool market based on loan size developed as investors rushed to buy mortgage pools offering prepayment protection.

The specified pool market based on loan size may be subdivided into four groups:

- **Low loan balance (LLB):** pools with maximum loan size less than \$85K;
- **Moderate loan balance (MLB):** pools with maximum loan sizes in the range of \$85K to \$110K;
- **High moderate loan balance (HLB):** pools with maximum loan sizes in the range of \$110K to \$150K; and,
- **Generic pools** (loan size greater than \$150K).³

Figure 2 shows the prepayment S-curves for different loan size (LS) buckets of 30-year agency mortgages based on prepayment speed data from 2006. At 100 bps of Refi incentive, LLB pools prepaid about 17% CPR slower than generic pools. When at-the-money (ATM) or slightly out-of-the money in terms of Refi incentive, LLB pools prepaid somewhat slower than generic pools. However, in a deep discount environment (say, at -100 bps Refi incentive), LLB pools actually prepaid 0.6%-1.5% CPR faster than generic pools. Thus, lower loan balance pools seem to show better convexity characteristics than generic pools.

In a premium environment, lower loan size pools prepay slower than higher loan size pools due to the weaker dollar incentive to refinance after accounting for the fixed cost of refinancing. When ATM or slightly out-of-the money, lower loan balance pools seem to prepay slower than generic pools because of larger percentage of cash-out refinancings on generic pools.

Although it is generally agreed upon that lower loan balance pools will prepay slower than generic pools when deep in-the-money, recent prepayment data show that lower loan size

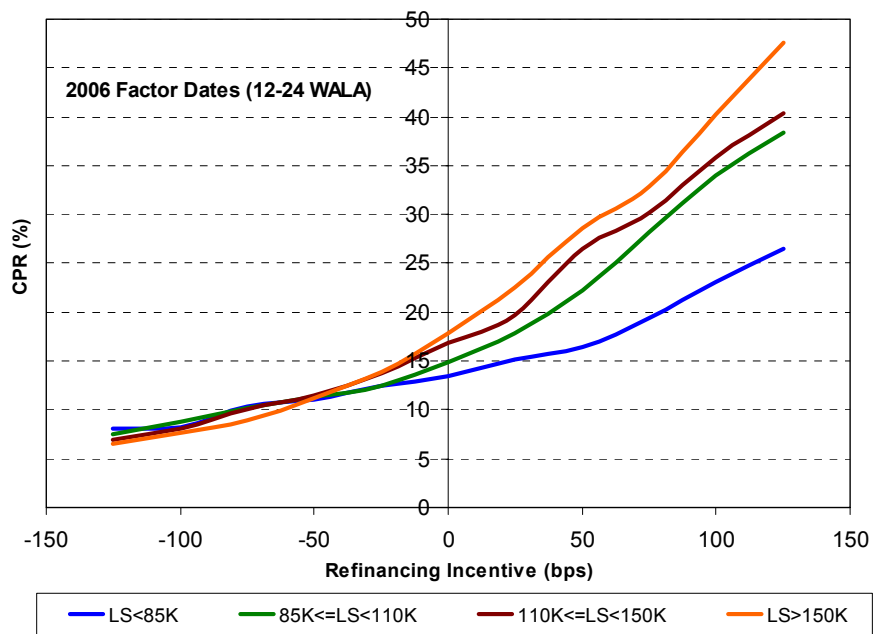
³ Recently, a new group for the maximum loan size range of \$150K - \$175K has been created. We will not consider it in the following discussion.

pools prepay faster than higher loan size pools when they deep out-of-the-money also. This is possibly due to weaker lock-in effect and because these borrowers are more likely to be first-time home buyers who are more likely to move soon. However, it is not clear if lower loan balance pools will continue to prepay faster than generic pools when they are deep out-of-the-money in a weak housing market.

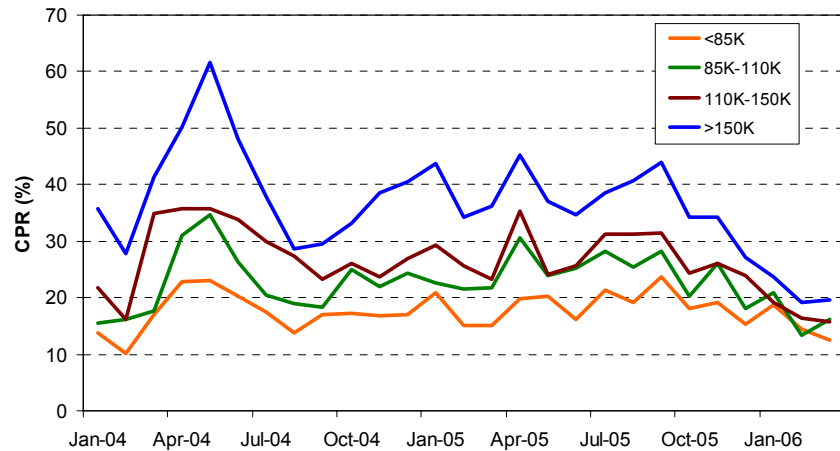
To further illustrate the prepayment protection offered by lower loan balance pools, we plot prepayment speeds of 12-24 WALA FNMA 6% pools since Jan'04 in Figure 3. These data show that LLB FN 6s prepaid at only 23% CPR when generic FN 6s prepaid at 61.5% CPR in April 2004 (May'04 factor date). In fact, prepayment protection progressively improves as the loan size declines from generic (>\$150K) to LLB pools. The valuation impact of these prepayment speed differentials can be quite significant. Let us look at a simple example to illustrate this.

In April 2004, FN 6s were super premiums trading at close to \$104. Thus, investors in FN 6s lost \$4 of premium on the realized prepayments of these pools in April 2004. For LLB FN 6s, a 23% CPR speed translates to a single monthly mortality (SMM) rate of 2.2%, while the 61.5% CPR speed on generic FN 6s translates to an SMM of 7.6%. The difference in prepayment speeds between LLB and TBA 6s translates to 6.9 ticks ($= (7.6\% - 2.2\%) * 4 * 32$) of difference in lost premium due to prepayments in one month alone.

Figure 2: Prepayment Response Curves of 30-year Agency MBS

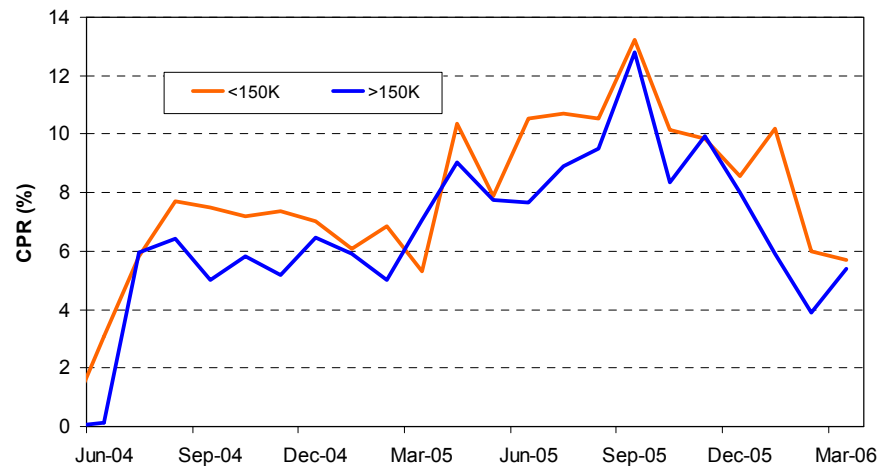


Source: Banc of America Securities

Figure 3: Prepayment Speeds of 12-24 WALA 30-year FNMA 6 Pools

Source: Banc of America Securities

Now, let's take a look at the historical prepayment speed data on 12-36 WALA FNMA 4.5s to get a sense for the better extension protection offered by lower loan balance pools. Figure 4 compares prepayment speeds on lower and higher loan size 30-year FNMA 4.5% coupon pools since June 2004. We divided the universe of 30-year FNMA 4.5s coupon into only two loan size buckets here because of the lack of sufficient volume for further sub-divisions. Clearly, lower loan size pools have prepaid faster than higher loan size pools with higher speed differentials occurring as the 4.5s coupon traded further below par.

Figure 4: Historical Prepayment Speeds of 12-36 WALA 30-year FNMA 4.5 Pools

Source: Banc of America Securities

Figure 5 shows theoretical pay-ups (equal OAS pay-ups) for different loan size pools across the 30-year coupon stack. The prepayment protection offered by LLB/MLB pools has been well recognized and the high market pay-ups for this collateral reflect it. More recently, the market has been paying attention to the extension protection offered by lower loan balance pools as well.

The pay-ups for lower loan balance pools depend on the following factors:

- **Magnitude of the TBA “premium”:** The higher the premium, the larger will be the pay-up for prepayment protection.
- **Dollar rolls:** Similar to pay-ups for any other specified pool characteristic, pay-ups for lower loan balance pools decline when dollar roll on the TBA of the underlying coupon is special.
- **Demand from the CMO market:** Recently, several investors are buying structured products (e.g. inverse IOs) backed by LLB premium collateral. This strong demand for derivative products structured from premium LLB pools drove their pay-ups close to or even slightly higher than their equal OAS pay-ups (Figure 5).

In general, pay-ups for lower loan balance pools are strongly dependent on the prepayment environment. Lower loan balance pools command significantly higher pay-ups in a Refi wave (e.g. 2002-2004), when the Media effect is strong and prepayment speeds on premium pools are very fast. Pay-ups on LLBs tend to shrink in a stable prepayment environment (2006 for example).

Figure 5: Relative Value Analysis for Lower Loan Size Pools

FNCL 5 (2 WALA, 5.70 GWAC, \$220K)			OAS = -7.8 bps			
Original Loan Size Limits	Average Loan Size	Price	Equal OAS Price Over TBA	LT Speed (Model)	Option Cost	Market Pay-up (ticks)
LS < 85K	75K	96-23	13	10.0	40.5	0
85K < LS < 110K	97K	96-19+	9+	9.7	41.2	0
110K < LS < 150K	130K	96-15+	5+	9.4	41.5	0
LS > 150K	205K	96-10		8.7	41.6	

FNCL 5.5 (2 WALA, 6.13 GWAC, \$230K)			OAS=-6.0 bps			
Original Loan Size Limits	Average Loan Size	Price	Equal OAS Price Over TBA	LT Speed (Model)	Option Cost	Market Pay-up (ticks)
LS < 85K	75K	99-06	16	11.6	49.2	4+
85K < LS < 110K	97K	99-01+	11+	11.5	51.3	1
110K < LS < 150K	130K	98-29	7	11.4	53.5	+
LS > 150K	230K	98-22		11.0	58.4	

FNCL 6.0 (7WALA, 6.62 GWAC, \$210K)			OAS = -7.3 bps			
Original Loan Size Limits	Average Loan Size	Price	Equal OAS Price Over TBA	LT Speed (Model)	Option Cost	Market Pay-up (ticks)
LS < 85K	75K	101-08	20	15.2	57.6	23
85K < LS < 110K	97K	101-02+	14+	15.8	61.0	10+
110K < LS < 150K	130K	100-28+	8+	16.7	64.5	3
LS > 150K	210K	100-20		19.0	68.9	

FNCL 6.5 (6 WALA, 7.15 GWAC, \$205K)			OAS = -16.7 bps			
Original Loan Size Limits	Average Loan Size	Price	Equal OAS Price Over TBA	LT Speed (Model)	Option Cost	Market Pay-up (ticks)
LS < 85K	75K	102-31	30	20.6	61.5	36
85K < LS < 110K	97K	102-22+	21+	22.6	62.7	19
110K < LS < 150K	130K	102-13	12	26.1	61.6	5
LS > 150K	205K	102-01		33.3	52.9	

As of 5/11/2007

Source: Banc of America Securities

Geographic Distribution

Figure 6 shows prepayment speeds organized by state for 30-year FNMA pools (with loan size > \$150K) in 2003 and 2006. The numbers under “Factor” give state level prepayment speeds as a fraction of the national aggregate speed. In the high prepayment speed environment that prevailed in 2003, the following large states were consistently faster: MI, MA, IL, WI, MO and CA. What is more interesting is that prepayment speeds in certain large states (NY, FL and TX) have been consistently slower than aggregate speeds in a Refi environment due to certain state regulations as discussed below:

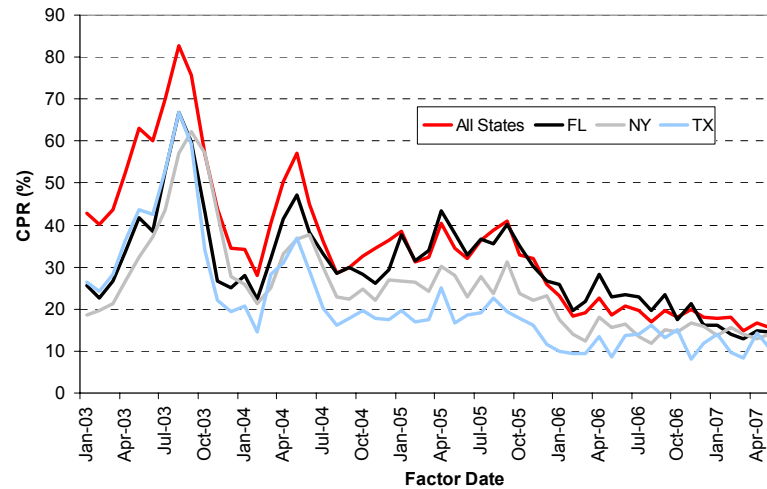
- **New York** – There is a state-wide mortgage recording tax as well as the New York City mortgage recording tax applicable to mortgages recorded in the City of New York. These additional transaction costs make mortgages originated in NY prepay slower than the US aggregate.
- **Florida** – Historically, prepayment speeds on FL pools were slower due to the mortgage tax of 35 bps in this state. However, note that FNMA pools from FL are actually prepaying faster than the national aggregate since 2004 because of the very strong HPA experienced by the state.
- **Texas** – There are some restrictions on cash-out refinancings for mortgages originated in Texas.

Figure 6: State Level Prepayment Speeds in 2003 and 2006 (with Loan Size > \$150K)

State/Territory	2003		2006		State/Territory	2003		2006	
	Speed	Factor	Speed	Factor		Speed	Factor	Speed	Factor
AK	25.9	0.71	8.1	0.79	NC	29.7	0.82	8.3	0.81
AL	22.3	0.61	8.5	0.83	ND	31.3	0.86	5.9	0.57
AR	24.4	0.67	7.4	0.72	NE	39.3	1.08	5.9	0.57
AZ	35.9	0.99	12.9	1.25	NH	47.9	1.32	9.0	0.87
CA	45.0	1.24	11.7	1.14	NJ	35.9	0.99	10.2	0.99
CO	39.4	1.09	8.4	0.82	NM	26.8	0.74	10.6	1.03
CT	35.7	0.98	8.7	0.84	NV	34.0	0.94	11.9	1.16
DC	37.4	1.03	10.0	0.97	NY	27.9	0.77	8.6	0.83
DE	33.5	0.92	7.5	0.73	OH	35.1	0.97	6.5	0.63
FL	24.7	0.68	10.8	1.05	OK	23.0	0.63	7.3	0.71
GA	28.3	0.78	8.2	0.80	OR	28.5	0.79	10.2	0.99
HI	29.7	0.82	10.4	1.01	PA	33.3	0.92	8.1	0.79
IA	33.7	0.93	5.8	0.56	PR	9.7	0.27	5.7	0.55
ID	26.8	0.74	10.6	1.03	RI	42.5	1.17	9.5	0.92
IL	49.1	1.35	11.4	1.11	SC	26.6	0.73	8.6	0.83
IN	35.1	0.97	7.5	0.73	SD	33.0	0.91	7.4	0.72
KS	36.8	1.01	6.9	0.67	TN	27.6	0.76	8.7	0.84
KY	37.0	1.02	7.6	0.74	TX	24.8	0.68	8.4	0.82
LA	25.2	0.69	15.3	1.49	UT	39.2	1.08	12.1	1.17
MA	53.8	1.48	9.4	0.91	VA	37.5	1.03	9.7	0.94
MD	40.1	1.10	11.4	1.11	VI	13.9	0.38	9.0	0.87
ME	38.2	1.05	9.0	0.87	VT	40.5	1.12	8.4	0.82
MI	49.7	1.37	7.5	0.73	WA	35.4	0.98	10.6	1.03
MN	37.5	1.03	7.4	0.72	WI	52.6	1.45	8.5	0.83
MO	46.4	1.28	9.4	0.91	WV	29.7	0.82	7.6	0.74
MS	23.1	0.64	11.2	1.09	WY	29.1	0.80	10.2	0.99
MT	28.1	0.77	8.6	0.83	US Agg	36.3		10.3	

Source: Banc of America Securities

Figure 7 compares prepayment speeds on FN 6% pools (12-24 WALA, Loan Size>\$150K) concentrated in NY, FL and TX with the national aggregate speeds for the same collateral. When prepayment speeds peaked in 2003, there was about a 20% CPR difference between the speeds on pools concentrated in slow states and the national aggregate but speed differences narrowed as overall prepayment speeds declined starting in 2004.

Figure 7: Prepayment Speeds of FN 6s (12-24 WALA, Loan Size>\$150K)

Source: Banc of America Securities

As a result of their slower prepayment speeds, mortgage pools concentrated with loans from slow prepayment states command a pay-up over TBAs for premium coupon pass-throughs. Although it is clear from the 2003 prepayment experience that premium mortgage pools concentrated in NY, FL and TX prepay substantially slower than the national aggregate in a strong Refi environment, there is some confusion regarding state-wide prepayment speeds of discount and slight premium pools. For discount and slight premium pools, other variables like home price appreciation (HPA), local economic conditions and the percentage of investor properties can sometimes dominate the prepayment speed differences arising from state regulations alone.

Leveraged Borrowers/Loans: Low FICO and High LTV Pools

Low FICO pools are supposed to offer both extension and prepayment protection versus high FICO pools. The general intuition behind this protection is as follows: Borrowers in low FICO pools are usually less efficient refinancers than borrowers in high FICO pools because of their lower credit quality. On the other hand, as the FICO score of a borrower improves over a period of time, they may find better rates which induce them to refinance even if the general interest rate environment has not changed (“credit curing”). Thus, low FICO pools are expected to prepay slower than generic pools in a refi environment and faster than generic pools in a discount environment.

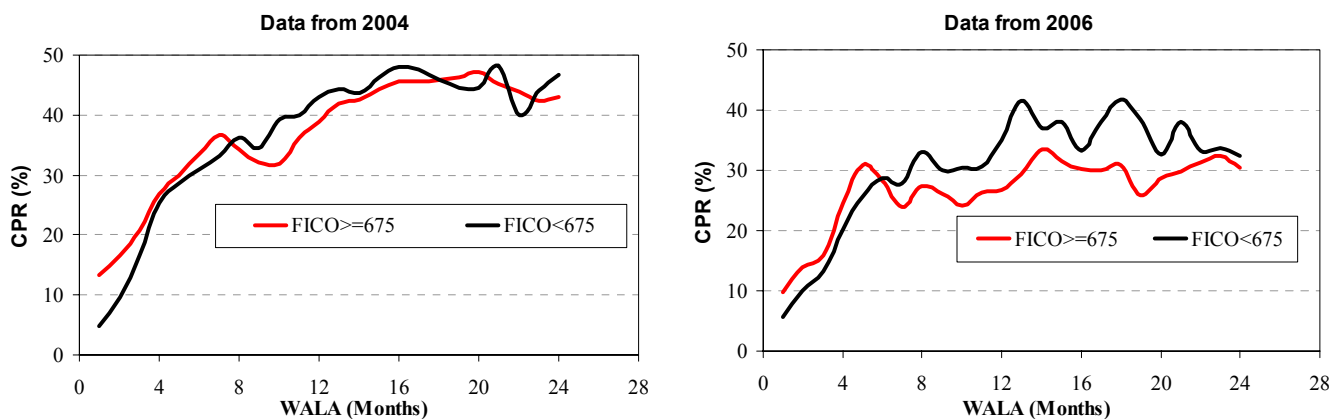
Figure 8 compares seasoning ramps for 75-125 bps in-the-money FNMA pools with FICO>675 and FICO<675 (loan size > \$120K) based on prepayment speeds observed in 2004 and 2006. In 2004, the low FICO pools started out by prepaying 6%-10% CPR slower but caught up with speeds on higher FICO pools by 8 WALA and have actually prepaid faster than higher FICO pools beyond 8 WALA. The seasoning ramp based on 2006 data shows that the prepayment advantage offered by low FICO pools has shortened further. In 2006, low FICO pools started prepaying faster than high FICO pools beyond only 6 WALA. One of the factors behind faster prepayment speeds of low FICO pools beyond 8-9 WALA in 2004 and 6-7 WALA in 2006 is apparently the higher GWACs of low FICO pools. In our data shown in Figure 8, low FICO pools have 10-15 bps higher GWAC than higher FICO pools. This higher GWAC appears to be offsetting the lower

credit quality of the borrowers in low FICO pools after the initial few months of mortgage origination.

It is also worth pointing out here that the prepayment differences between lower and higher FICO pools are most noticeable during the first few months after a mortgage pool is originated. As the pools season and borrowers' credit scores improve, the prepayment protection offered by low FICO pools goes away. In fact, pools backed by lower FICO borrowers can actually prepay faster after the initial seasoning period as shown in Figure 9, which plots the prepayment response curves of 12-24 WALA pools in 2004 and 2006.

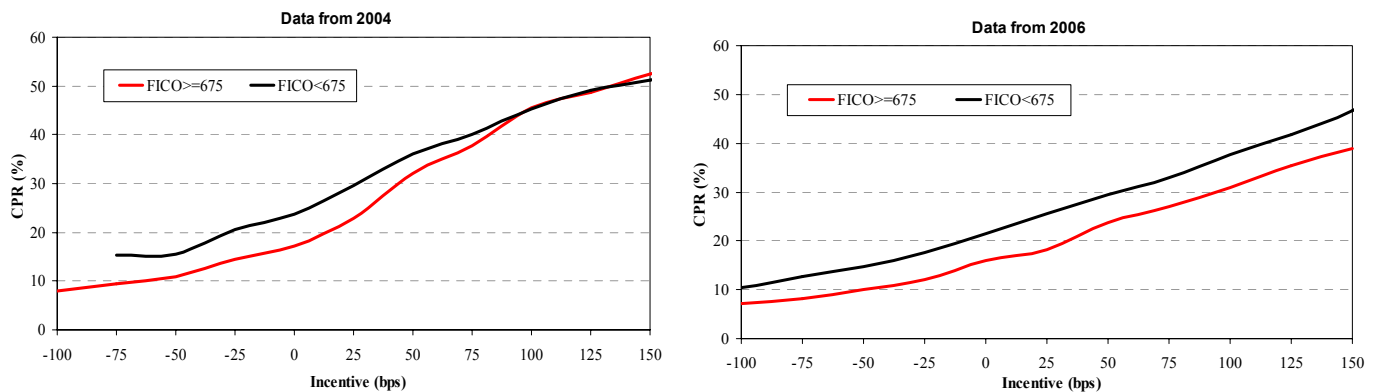
In 2004, low FICO pools that were deep ITM prepaid at similar speeds as high FICO pools, but at other incentive levels low FICO pools were faster. In 2006, low FICO pools prepaid faster than high FICO pools at all incentive levels. Noting that the data in Figure 9 is for 12-24 WALA pools, it appears that low FICO pools are completely "credit cured" by 12-24 WALA and are prepaying faster.

Figure 8: FICO Effect - Prepayment Ramps for 75-125 bps ITM (Loan Size > \$120K)



Source: Banc of America Securities

Figure 9: FICO Effect - S-Curves from 2004 and 2006 (12-24 WALA, Loan Size > \$120K)



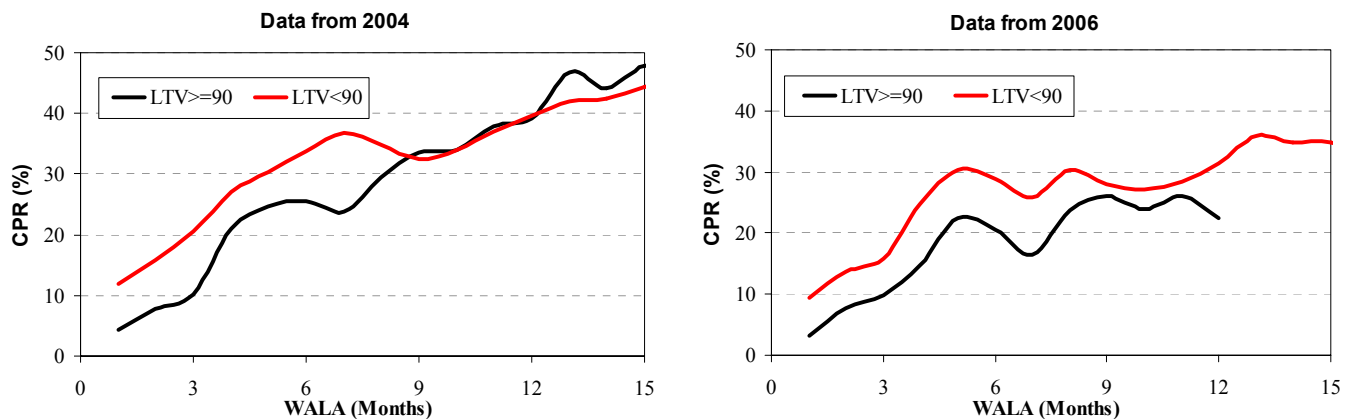
Source: Banc of America Securities

A similar line of reasoning applies to high loan-to-value (LTV) pools which are supposed to offer both extension and call protection versus lower LTV pools. Borrowers in high LTV pools are usually less efficient refinancers than borrowers in low LTV pools because

of their higher leverage. However, as the LTV of a mortgage declines over a period of time due to home price appreciation or principal pay-downs, borrowers may find better rates that induce them to refinance.

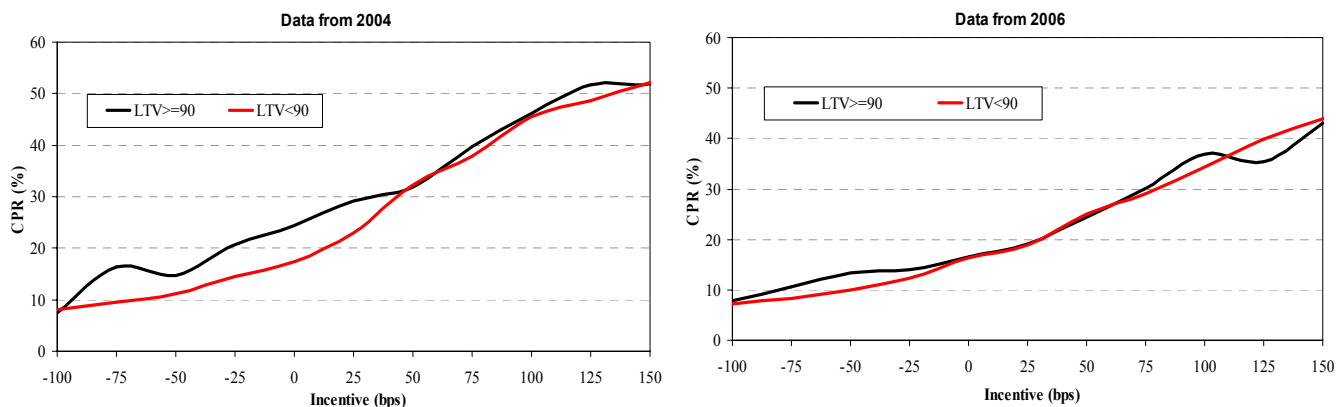
The recent prepayment experience on 75-125 bps ITM pools shows that high LTV pools offer prepayment protection for only 8-9 months (Figure 10). Here we should point out that the prepayment speed information in Figure 10 for 2006 beyond 11-12 WALA is not reliable as the data is sparse. The prepayment response curves for 12-24 WALA pools (Loan size > \$120K) shown in Figure 11 demonstrate that high LTV loans have been prepaying somewhat faster than low LTV pools when out-of-the-money, and the two groups are prepaying similarly when in-the-money.

Figure 10: LTV Effect - Prepayment Ramps for 75-125 bps ITM (LS > \$120K)



Source: Banc of America Securities

Figure 11: LTV Effect - S-Curves from 2004 and 2006 (12-24 WALA, LS > \$120K)



Source: Banc of America Securities

Our assessment of low FICO and high LTV pools is that the level of call and extension protection they provide depends on several factors. For instance, the charts in Figure 11 show that high LTV pools provide excellent extension protection, but it is not clear if the

same protection would be available in a strongly negative HPA scenario. Similarly, at the peak of the Refi wave, both low FICO and high LTV pools offered significant prepayment protection but this protection withered away as we came out of the Refi wave and low FICO pools are now actually prepaying faster beyond 8 WALA when in-the-money. Consequently, there is not much conviction on the prepayment protection offered by low FICO and high LTV pools in the current environment. We expect low FICO and high LTV pools to offer meaningful prepayment protection when in-the-money if the housing market were to remain very weak but the prepayment protection they offer is questionable in a strong housing market. In a strong housing market, these characteristics are likely to offer prepayment protection only if there is another massive Refi wave.

Other Specified Pool Characteristics

In addition to the four collateral characteristics discussed above, the following characteristics of mortgage pools are also actively priced in the specified pool markets.

- **Low GWAC Pools:** Typically, mortgage pools will have GWACs that are approximately 50 bps higher than net coupon. All else being equal, TBA delivered pools should have somewhat higher GWACs than the average GWAC for the cohort for current coupon and premium TBAs. Historically, higher GWAC pools have prepaid faster than corresponding cohorts when they are premiums and hence low GWAC pools command a pay-up over TBAs for premium securities. The significance of GWACs declines as we move from current coupon securities towards deep discounts or deep premiums.
- **New Origination (low WALA) Premium Pools:** We have discussed this topic before under seasoning but premium pools are usually talked about in terms of new origination (low WALA) pools rather than as seasoned pools. The new origination premium pools command a premium because they prepay slower than generic collateral during the Refi ramp.
- **Fixed-Rate Interest-Only (IO) versus Amortizing Pools:** From its near non-existent status prior to 2H'2005, fixed-rate IOs have become an important component of the specified pools sector in the agency pass-through market. Please see our primer titled "*Fixed-rate IO Mortgages*" for more details on this topic. We note here that the fixed-rate IOs are different from other specified pools in the agency pass-through market in that they are not eligible for TBA delivery.

III. Summary

The significance of the specified pool market has grown rapidly over the past 4-5 years. This market is most active for seasoned and lower loan balance collateral at the moment, but premium pools from states that are known for slow prepayment speeds and pools with low FICO or high LTVs also command a significant premium at times of heavy refinancing activity.

Most of our discussion regarding the fair value of specified pool collateral has been in terms of the equal OAS pay-up. i.e., the pay-up a specified pool should command relative to TBA with the same coupon such that the specified pool and the TBA have equal OASs. In general, specified pools rarely trade at more than 50% of the theoretical pay-up because of their lower liquidity relative to TBAs and the funding advantage offered by dollar rolls.

Finally, there are a few important points to be noted about specified pool valuations:

- Specified pools have durations, curve exposures and convexities that are different from the corresponding TBA pools. OAS calculations account for these differences but market participants seem to frequently ignore them for hedging purposes. This can potentially lead to heavy losses or gains when interest rates move substantially.
- An important aspect of the specified pool market is that all pay-ups decline substantially when dollar roll of the corresponding coupon TBA trades special.
- Demand from CMO desks plays an important role in terms of determining pay-ups for specified pools.

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